

Project – Low Level Design

on

Healthcare Content Generator

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement

In the healthcare industry, professionals are required to create large amounts of documentation including patient summaries, clinical reports, treatment notes, and medical explanations. Preparing these documents manually can take a considerable amount of time and effort. Additionally, maintaining consistency in formatting, tone, and medical terminology can be challenging, especially when documentation is created by different individuals. This often leads to inefficiencies and increases the workload of healthcare professionals. Moreover, incorrect or inconsistent documentation can affect communication among medical staff and may reduce the clarity of patient records. Therefore, there is a need for an intelligent system that can assist healthcare professionals in generating high-quality medical content quickly while ensuring accuracy and consistency. The Healthcare Content Generator addresses this problem by using Generative AI and prompt engineering to automate the process of content generation while maintaining professional healthcare standards.

2. Introduction

The Healthcare Content Generator is a specialized Generative AI application designed to assist healthcare professionals in creating accurate, well-structured, and professional medical documentation. In the healthcare sector, doctors, nurses, and medical staff often spend a significant amount of time preparing reports, patient summaries, clinical notes, and other documentation. This process can be time-consuming and may sometimes lead to inconsistencies in formatting or terminology. The proposed system uses advanced Generative AI techniques and prompt engineering to convert simple user inputs or medical topics into high-quality professional content. By integrating powerful language models through an LLM API, the system can generate structured and medically appropriate responses. The application also ensures that the generated content follows standard medical terminology and professional tone. As a result, the Healthcare Content Generator reduces manual effort, saves time, and improves the efficiency of healthcare professionals while maintaining clarity and consistency in medical documentation.

3. System Overview

The Healthcare Content Generator is designed as an intelligent AI-powered system that automatically generates healthcare-related content based on user input. The system works by accepting a simple prompt or topic from the user, such as a patient condition or medical procedure. This input is processed using carefully designed prompt engineering techniques that guide the AI model to generate structured and meaningful output. The system also integrates a vector database to store and retrieve relevant medical knowledge, guidelines, and reference content. When a user submits a request, the system retrieves relevant information from the vector database and combines it with the prompt before sending it to the language model through an API. The language model then generates detailed and professional healthcare content such as patient summaries, medical explanations, or documentation. The final output is displayed to the user in a structured and readable format. Overall, the system improves productivity and helps healthcare professionals create high-quality documentation quickly and efficiently.

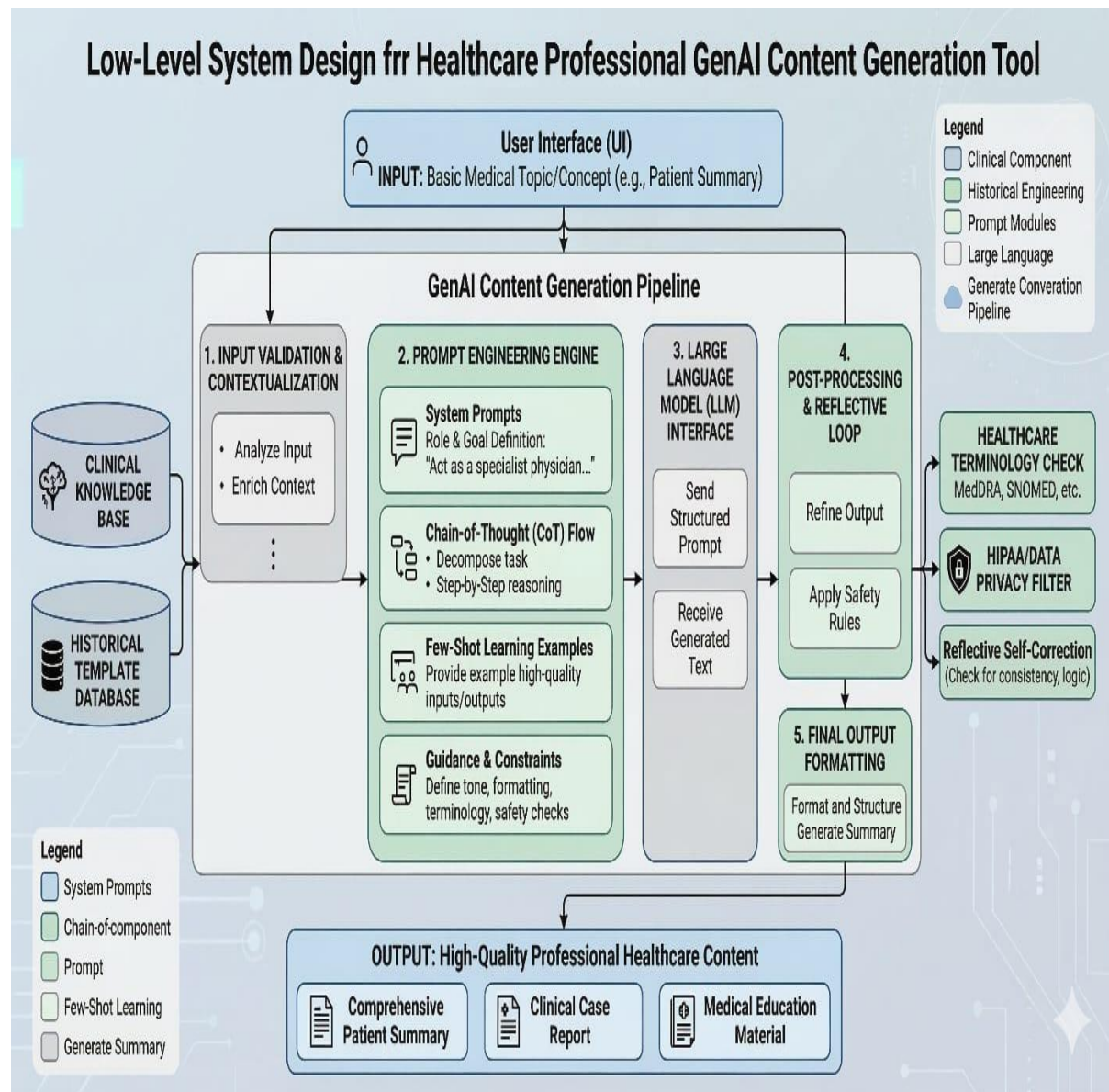
4. Tools and Technologies Used

Tool / Technology	Purpose
Python	Backend development and AI integration
LLM API	Generating AI-based content
Vector Database	Storing embedding for similarity search
Prompt Engineering	Improving AI responses
REST APIs	Communication between modules

5. System Architecture Diagram

The **Healthcare Content Generator** uses **Python**, a **Vector Database**, and an **LLM API** to generate healthcare content. First, the user enters a topic or request in the system. The Python backend processes the input and uses **prompt engineering** to create a structured prompt. Then, the system retrieves relevant

information from the **vector database** to provide context. This prompt and context are sent to the **LLM API**, which generates professional healthcare content such as patient summaries. Finally, the generated output is formatted and displayed to the user.



6. Module Design

6.1. User Management Module

Handles user-related activities.

Functions:

- User registration
- Login authentication
- Profile management

6.2. Prompt Processing Module

This module improves the user input before sending it to the AI model.

Functions:

- Prompt formatting
- Context addition
- Template selection

6.3. Content Generation Module

Responsible for generating healthcare content using the LLM API.

Functions:

- Send prompts to AI model
- Receive generated text
- Validate output

6.4. Vector Search Module

This module stores and retrieves embeddings using the vector database.

Functions:

- Store prompt embedding
- Perform similarity search
- Retrieve relevant content
-

6.5. Content Storage Module

Stores generated reports and user data.

Functions:

- Save generated content
- Retrieve previous reports
- Manage document history

7.DataBase Design

7.1. Users Table

Field Name	Data Type	Description
user_id	Integer (Primary Key)	Unique ID for each user
name	Varchar	Name of the healthcare professional
email	Varchar	User email address
role	Varchar	Role of the user (Doctor, Nurse, Admin)

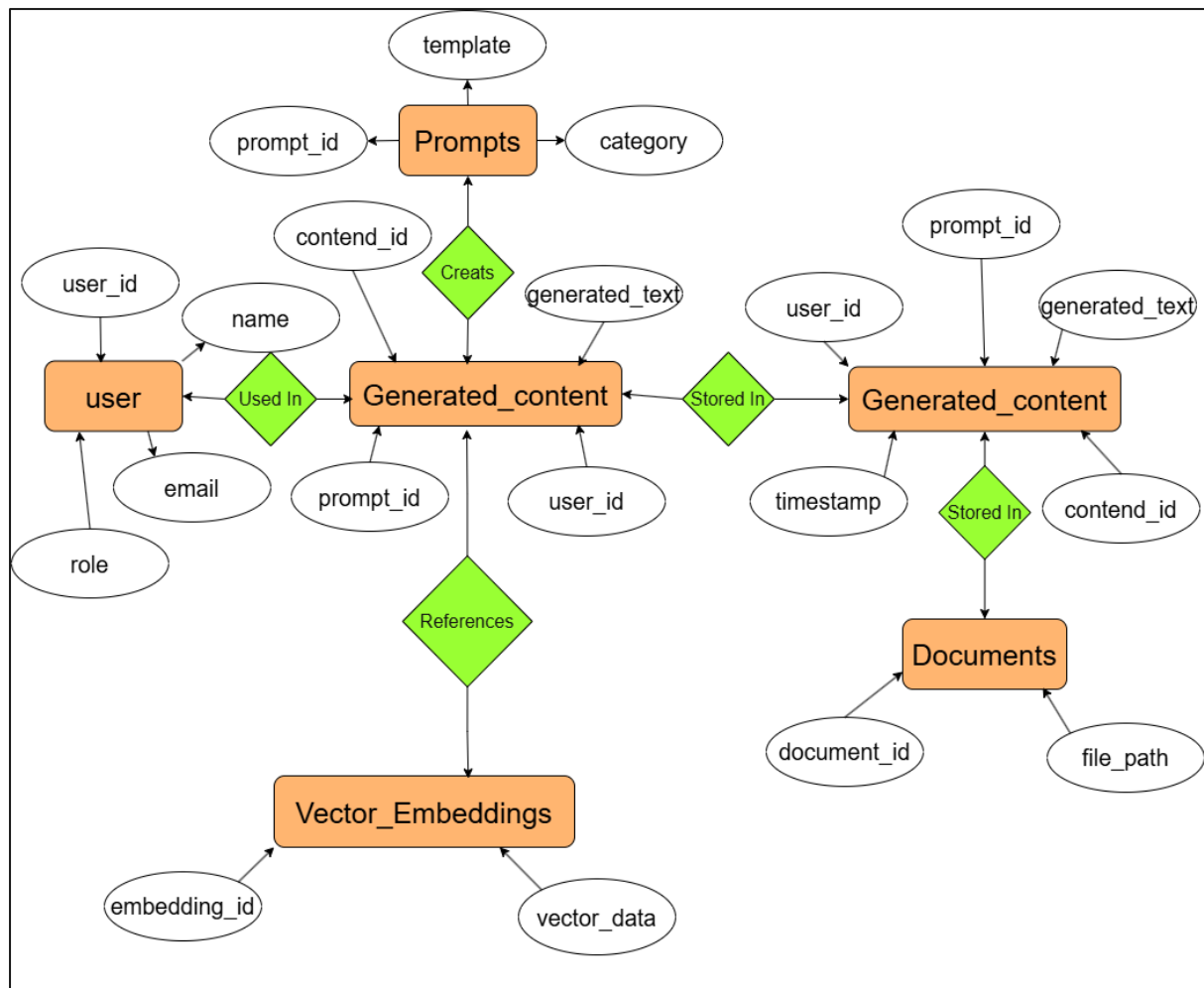
7.2. Prompt Processing Module

Field Name	Data Type	Description
prompt_id	Integer (Primary Key)	Unique ID for each prompt template
template	Text	Prompt template used for content generation
category	Varchar	Category of prompt (Patient Summary, Case Report, etc.)

7.3. Generated Content Table

Field Name	Data Type	Description
content_id	Integer (Primary Key)	Unique ID for generated content
user_id	Integer (Foreign Key)	ID of the user who generated the content
prompt_id	Integer (Foreign Key)	Prompt template used for generation
generated_text	Text	AI-generated healthcare content
timestamp	DateTime	Time when the content was generated

8.ER Diagram



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